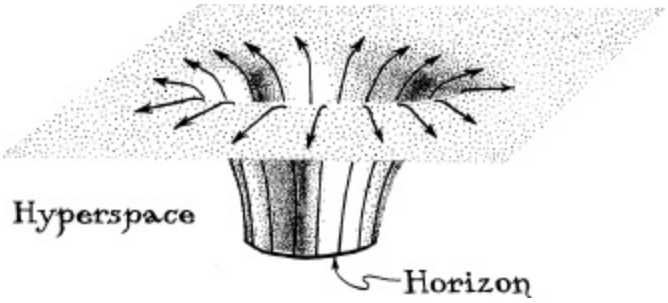
after the golden age ended, came a traumatic, explosive split in which almost everybody in Zel'dovich's team went one way and he went another, psychologically wounded but free from encumbrances, free to begin building afresh. Sadly, his rebuilding would not be successful. Never again would he surround himself with a team so talented and powerful as that which he, with Novikov's assistance, had led. But Novikov, now an independent researcher, would come into his own in the 1980s as the talented leader of a reconstructed team.

We can depict the curvature of space around a charged black hole, and the hole's electric field lines, using an embedding diagram (left half of Figure 7.6). This diagram is essentially the same as the one in the lower right of Figure 3.4, but with the star (black portion of Figure 3.4) removed because the star is inside the black hole and thus no longer has contact with the external Universe. Stated more carefully, this diagram depicts the equatorial "plane"—a two-dimensional piece of the hole's space—outside the black hole, embedded in a flat, three-dimensional hyperspace. (For a discussion of the meaning of such diagrams, see Figure 3.3 and the accompanying text.) The equatorial "plane" is cut off at the hole's horizon, so we are seeing only the hole's exterior, not its interior. The horizon, which in reality is the surface of a sphere, looks like a circle in the diagram because we are seeing only its equator. The diagram shows the hole's electric field lines sticking radially out of the horizon. If we look down on the diagram from above (right side of Figure 7.6), then we do not see the curvature of space, but we do see the electric field lines more clearly.



7.6 Electric field lines emerging from the horizon of an electrically charged black hole. *Left:* Embedding diagram. *Right:* View of the embedding diagram from above.